

Ideation Phase

Brainstorm & Idea Prioritization

Field	Details
Date	18 June 2026
Team ID	24EG105H22
Project Name	HouseHunt – House Rent Management System
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement

Finding rental houses through traditional methods such as brokers, newspaper advertisements, social media posts, and personal references is time-consuming and inefficient. Tenants often struggle to find accurate property information, while property owners face difficulties in reaching potential tenants and managing property listings effectively. There is a need for a centralized platform that simplifies property searching, property listing, and rental management through a secure and user-friendly system.

Step-2: Brainstorm, Idea Listing and Grouping

List of Ideas

#	Idea	Description
1	Online Property Listing Portal	Allow owners to post rental properties with photos, price, and details
2	Role-Based Access Control	Separate dashboards for Tenants, Owners, and Admins with different permissions
3	Admin Approval Workflow	Properties go live only after admin verifies and approves them
4	Smart Search & Filter	Filter by location, property type, price range, and number of bedrooms
5	Booking Request System	Tenants send booking requests; owners accept or reject with reason
6	JWT Authentication	Secure token-based login without sessions, scalable across services
7	Property Detail Pages	Detailed view with images, amenities, owner contact, and booking CTA
8	Owner Dashboard	Owners see all their listings and manage incoming booking requests

#	Idea	Description
9	Renter Dashboard	Renters track their submitted booking requests and statuses
10	Admin Analytics Panel	Platform-wide statistics: total properties, users, bookings, pending approvals

Grouping of Ideas

Group / Theme	Ideas Included
User Management	Role-Based Access Control, JWT Authentication
Property Management	Online Property Listing Portal, Property Detail Pages, Smart Search & Filter
Booking Management	Booking Request System, Owner Dashboard, Renter Dashboard
Administration	Admin Approval Workflow, Admin Analytics Panel

Brainstorming MindMap (Summary)

Central Node: HouseHunt – House Rent Management System

Branch	Sub-nodes
Users	Tenant → Search → Book → Track Owner → List → Manage → Approve Admin → Approve → Monitor
Features	Search/Filter Booking Workflow Admin Approval JWT Auth Dashboards
Technology	React.js Express.js PostgreSQL Drizzle ORM JWT Tailwind CSS
Impact	No Broker Fee Verified Listings Fast Search Transparent Booking

Step-3: Idea Prioritization

Prioritization Table (Impact vs Effort Matrix)

Idea	Impact (1-5)	Effort (1-5)	Priority Score (Impact/Effort)	Priority
JWT Authentication	5	2	2.5	High
Role-Based Access Control	5	2	2.5	High
Online Property Listing Portal	5	3	1.67	High
Admin Approval Workflow	4	2	2.0	High

Idea	Impact (1-5)	Effort (1-5)	Priority Score (Impact/Effort)	Priority
Smart Search & Filter	5	3	1.67	High
Booking Request System	5	3	1.67	High
Owner Dashboard	4	2	2.0	High
Renter Dashboard	4	2	2.0	High
Admin Analytics Panel	4	3	1.33	Medium
Property Detail Pages	4	2	2.0	High

Final Selected Idea

HouseHunt – House Rent Management System

Reason for Selection

HouseHunt was selected because it addresses a real-world problem faced by tenants and property owners. The project is technically feasible using the MERN/PERN Stack and provides opportunities to implement modern web technologies such as React.js, Node.js, Express.js, PostgreSQL, JWT Authentication, CRUD Operations, and Role-Based Access Control. The system offers practical value and has significant scope for future enhancements.